Trigonometric Substitution (7.3)

$$\int \sqrt{16-x^2}\,dx$$

$$\int x\sqrt{16-x^2}\,dx$$

$$\int_a^b \sqrt{16-x^2}\, dx$$

$$\int \frac{1}{\sqrt{1-x^2}}\,dx$$

$$\int \frac{1}{1+x^2}\,dx$$

"Rules of thumb" for trigonometric substitution.

$\sqrt{a^2-x^2}$	$x = a\sin\theta \quad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$
$\sqrt{a^2+x^2}$	$x = a\tan\theta \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$
$\sqrt{x^2-a^2}$	$x = a\sec\theta \quad 0 \le \theta < \frac{\pi}{2}$ or $\pi \le \theta < \frac{3\pi}{2}$

$$\int \sqrt{16-x^2}\, dx$$

$$\int \frac{1}{x^2\sqrt{x^2+9}}\,dx$$

$$\int_0^2 \frac{1}{(x^2+4)^{3/2}}\,dx$$

$$\int_0^{2/3} \frac{1}{\sqrt{4-9x^2}}\,dx$$

$$\int \frac{1}{\sqrt{x^2+x}}\,dx$$